A Frozen NCP deployment in three paired-chamber AF cohorts

Frozen Fig 1 definitions were applied unchanged across cohorts; no cohort-specific re-selection or recalibration.

INPUT

Frozen NCP from Fig 1

30 programs → 16 families

leading-edge gene sets
canonical direction s_p
reference magnitude M_p
family topology and display order

COHORTS

Three paired LA–RA cohorts

GSE128188	5 SR	5 AF
GSE79768	6 SR	7 AF
GSE115574	15 SR	13 AF

51 paired donors
26 SR | 25 AF
One complete LA–RA pair per donor

QUANTIFICATION

Donor-level chamber quantification

$$aCAIrel_{d,p} = \frac{s_p\left(S^{LA}_{d,p} - S^{RA}_{d,p}\right)}{M_p}$$

$$aCAIrel_{d,f} = \frac{1}{|P_f|}\sum_{p \in P_f} aCAIrel_{d,p}$$

- Same singscore implementation
- M_p fixed from GSE112339
- Equal-weight program-first aggregation
- No disease-cohort re-selection or recalibration

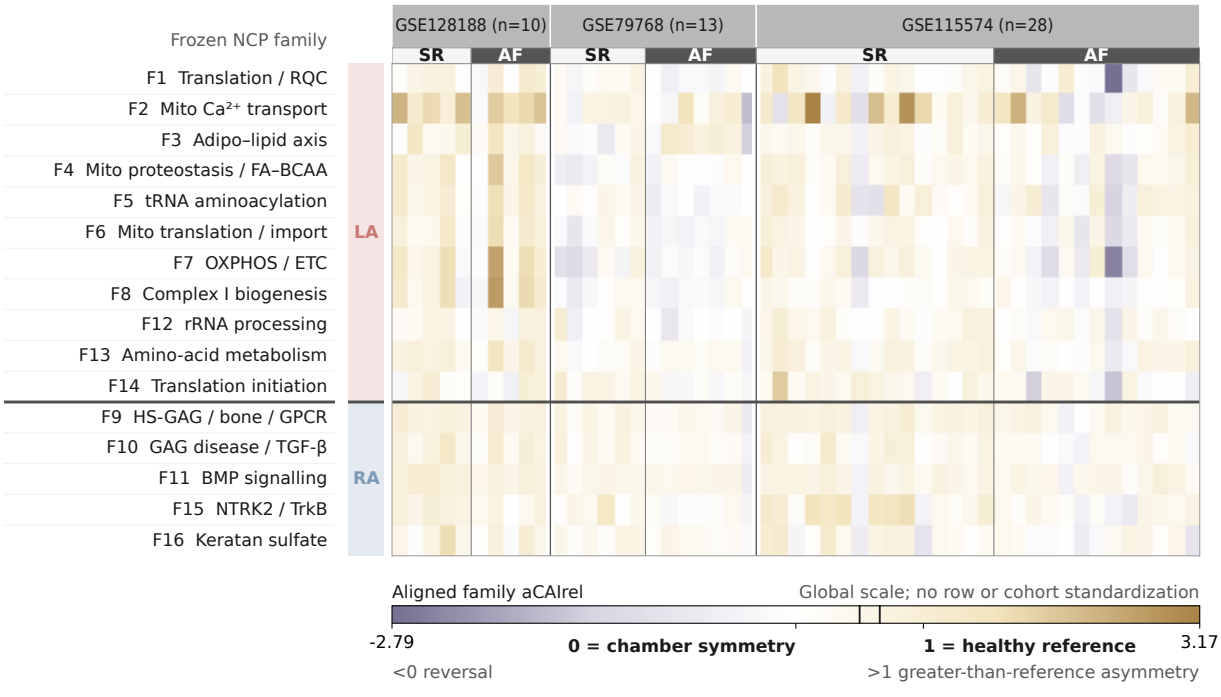
OUTPUTS

- **B** Donor-level family aCAIrel
- **C** Cohort-specific rewiring states
- **D** AF–SR effects and consistency

B Donor-level family aCAIrel across three paired-chamber bulk cohorts

16 frozen NCP families · 51 donors with complete LA–RA pairs · 26 SR · 25 AF

Columns: cohort → rhythm → donor



C Cohort-specific family rewiring states

16 frozen NCP families × 3 cohorts = 48 descriptive state calls

Descriptive AF-relative-to-SR states; not significance tests.

	ID	Frozen NCP family	GSE128188 5 SR / 5 AF	GSE79768 6 SR / 7 AF	GSE115574 15 SR / 13 AF
LA	F1	Translation / RQC	Amp	Rev	Rev
LA	F2	Mito Ca²⁺ transport	Pres	Att	Att
LA	F3	Adipo–lipid axis	Pres	Amp	Pres
LA	F4	Mito proteostasis / FA–BCAA	Pres	Pres	Att
LA	F5	tRNA aminoacylation	Pres	Rev	Att
LA	F6	Mito translation / import	Amp	Rev	Rev
LA	F7	OXPHOS / ETC	Amp	Rev	Rev
LA	F8	Complex I biogenesis	Amp	Rev	Rev
LA	F12	rRNA processing	Pres	Rev	Att
LA	F13	Amino-acid metabolism	Pres	Pres	Att
LA	F14	Translation initiation	Pres	Rev	Rev
RA	F9	HS-GAG / bone / GPCR	Pres	Att	Att
RA	F10	GAG disease / TGF-β	Pres	Att	Att
RA	F11	BMP signalling	Pres	Pres	Pres
RA	F15	NTRK2 / TrkB	Att	Att	Att
RA	F16	Keratan sulfate	Pres	Att	Att

Amplified (n=5) · Preserved (n=16) · Attenuated (n=15) · Reversed (n=12)

corner wedge: SR-reference floor applied (5/48 calls)

D Cross-cohort AF-SR effects and consistency

Cohort-specific and pooled Hedges' g across 16 frozen NCP families

